A Domain Language for Specifying Controlled Methods

A controlled-method specification pitch for reproducible research teams

Audience and failure mode

Methods are interfaces

- A method is an interface between a question, its inputs, and the claims its outputs can support.
- When that interface is implicit, later changes can silently alter the meaning of a result.

A small controlled language

- The DSL names stages, dependencies, units, and exports as data that can be parsed and validated.
- Compilation remains deterministic and offline by default, with no hidden network or LLM path.

Source: projects/templates/template_methods_paper/src

Negative controls carry the argument

- Malformed stages, changed literals, and invalid units must fail before a receipt can be treated as current.
- The focused tests exercise those failures without repeating the full refresh path.

Source: projects/templates/template_methods_paper/tests

What the receipt means

- It binds a local source revision, method plan, output schema, and deterministic compiler result.
- It does not substitute for a domain comparator, real data provenance, or owner-authorized publication.

Source: projects/templates/template_methods_paper/README.md

A bounded next step

- Encode one production method and review the diff with its domain owner.
- Promote only the claims whose evidence and acceptance predicates are explicit.